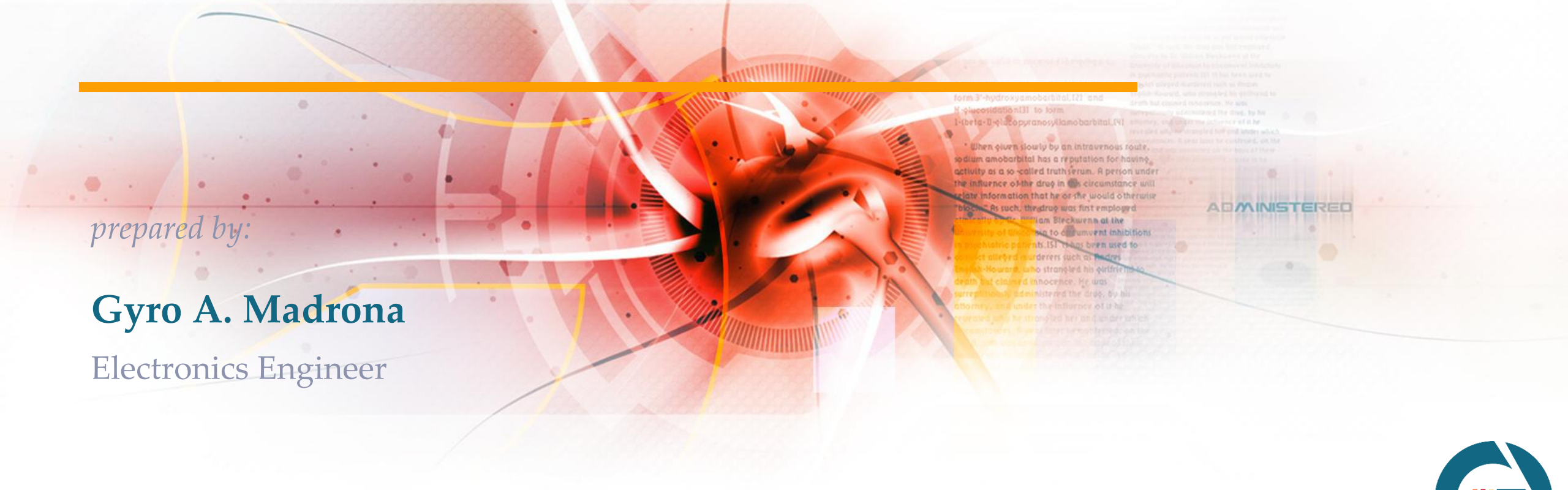
THE ELLIPSE

CONICAL SECTIONS



prepared by:

Gyro A. Madrona

Electronics Engineer

GRADING SYSTEM

FINAL GRADE

30%

PRELIM

30%

MIDTERM

40%

ENDTERM

TERM GRADE

40%

WEEKLY

60%

EXAM

PENALTY

0

ABSENT

3

45-min LATE

5 max

15-min LATE

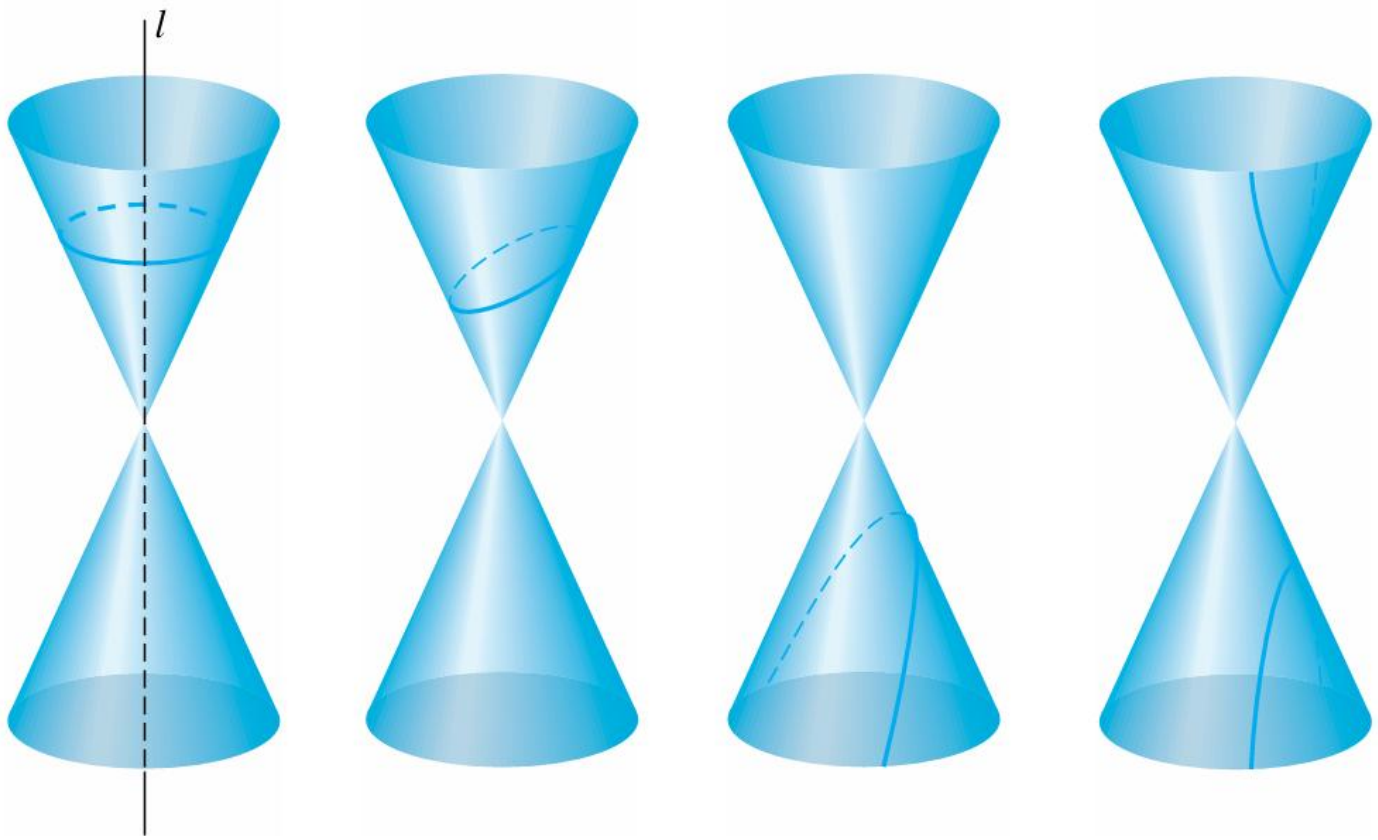
TOPIC OUTLINE

Ellipse with Center at the Origin

Ellipse with Center at $V(h, k)$

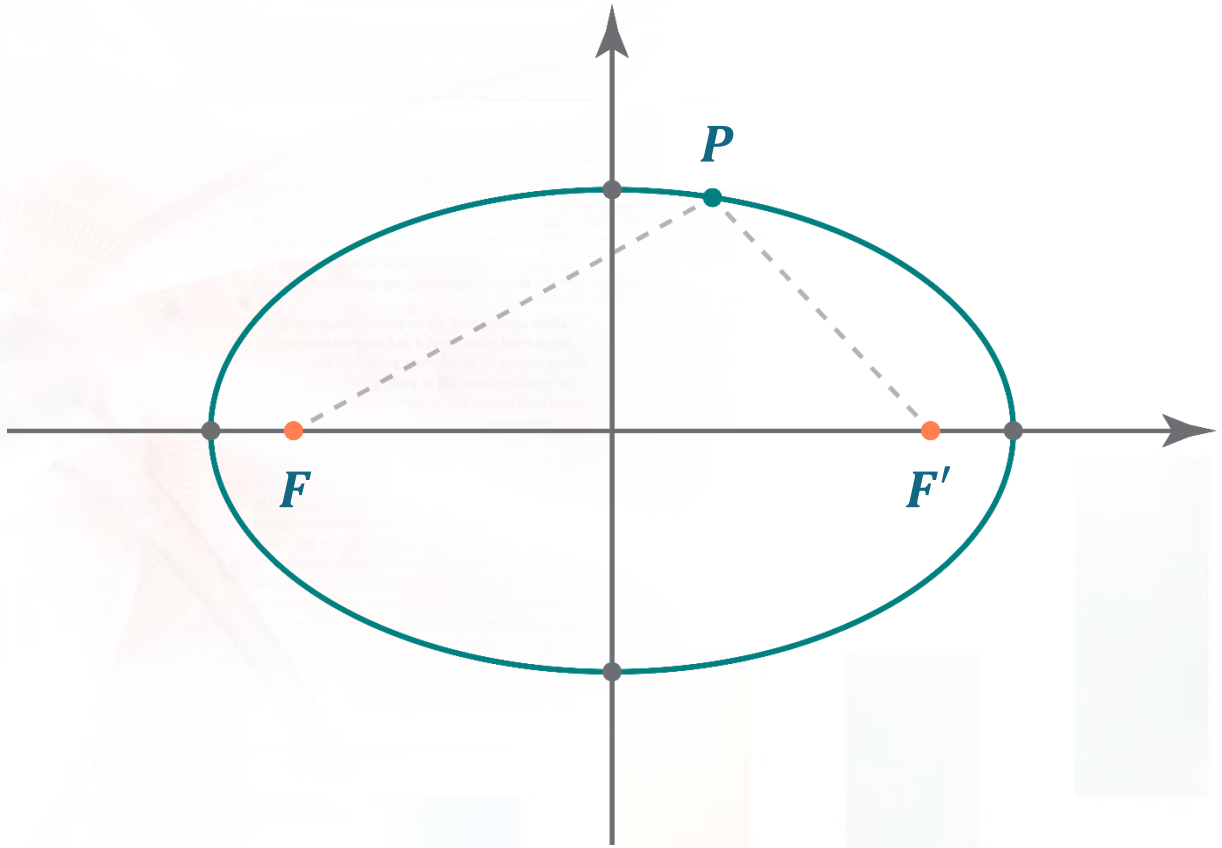
ELLIPSE WITH CENTER AT THE ORIGIN

CONICAL SECTIONS



Circle | **Ellipse** | Parabola | Hyperbola

An **ellipse** is the set of all points in a plane, the **sum** of whose **distances** from two fixed points (the **foci**) in the plane is a positive **constant** ($2a$)

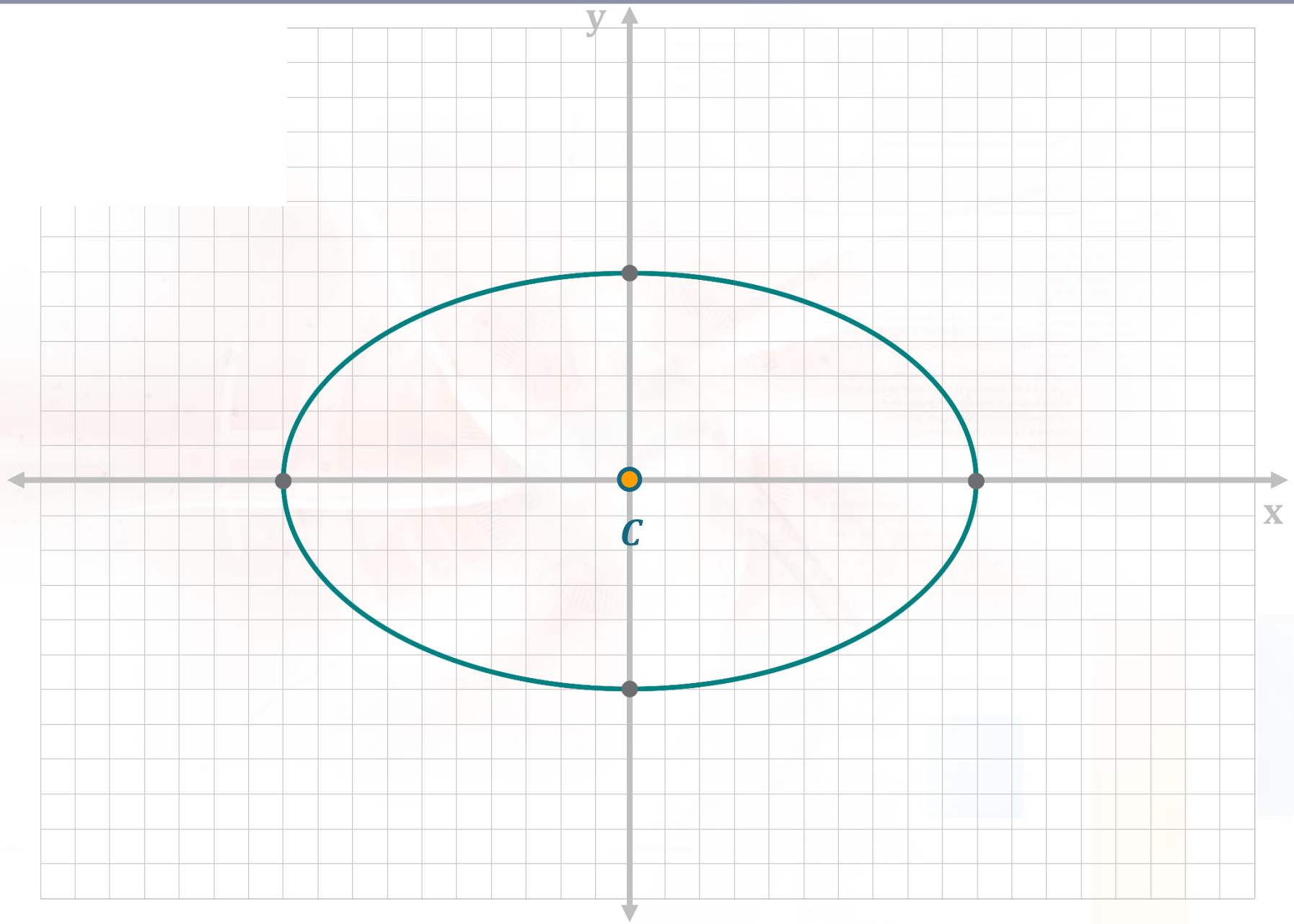


$$d(P, F) + d(P, F') = 2a$$

CENTER AT THE ORIGIN

center

$C(0, 0)$



VERTICES

vertices

$V(\pm a, 0)$

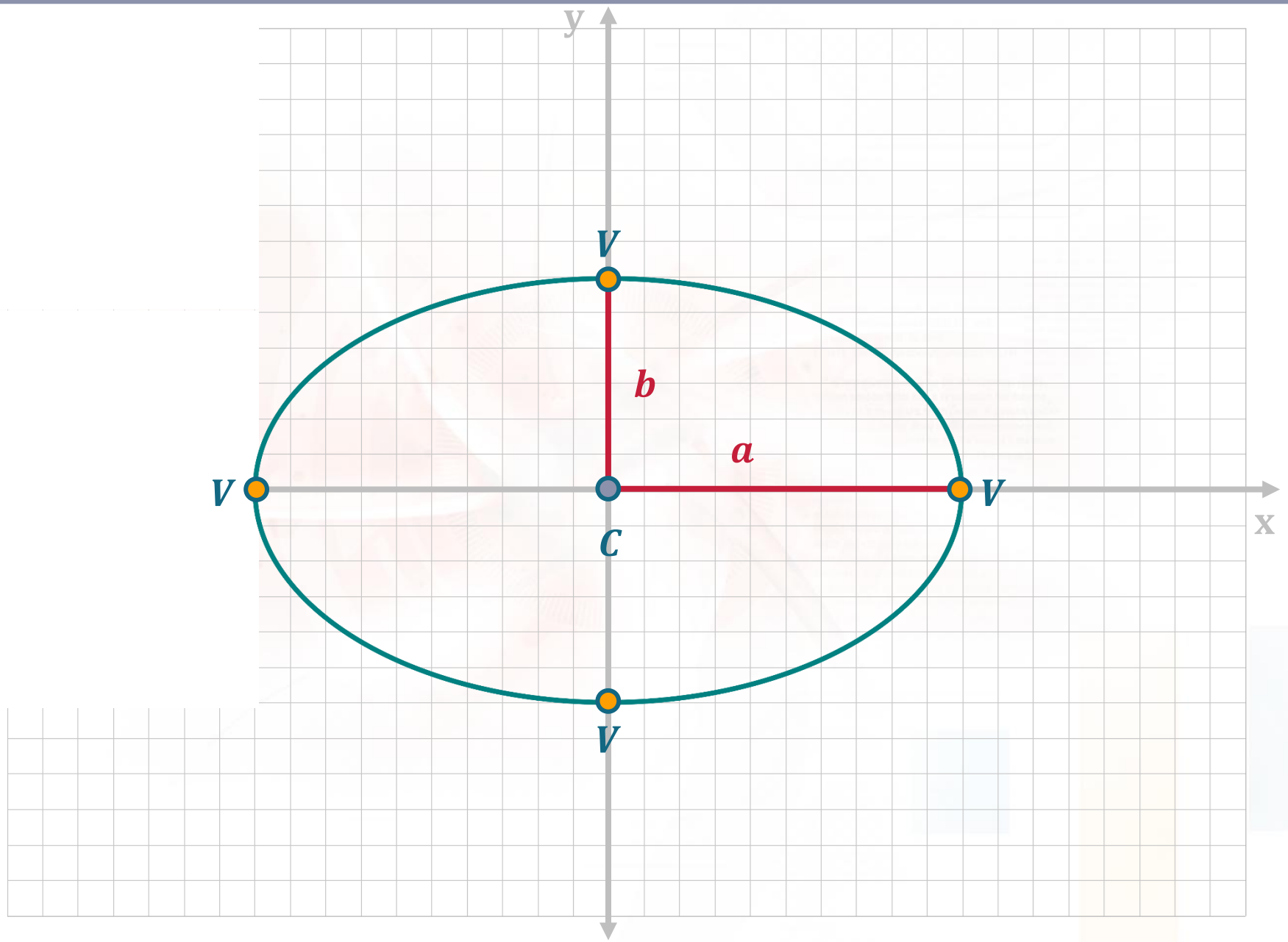
$V(0, \pm b)$

major axis

$2a$

minor axis

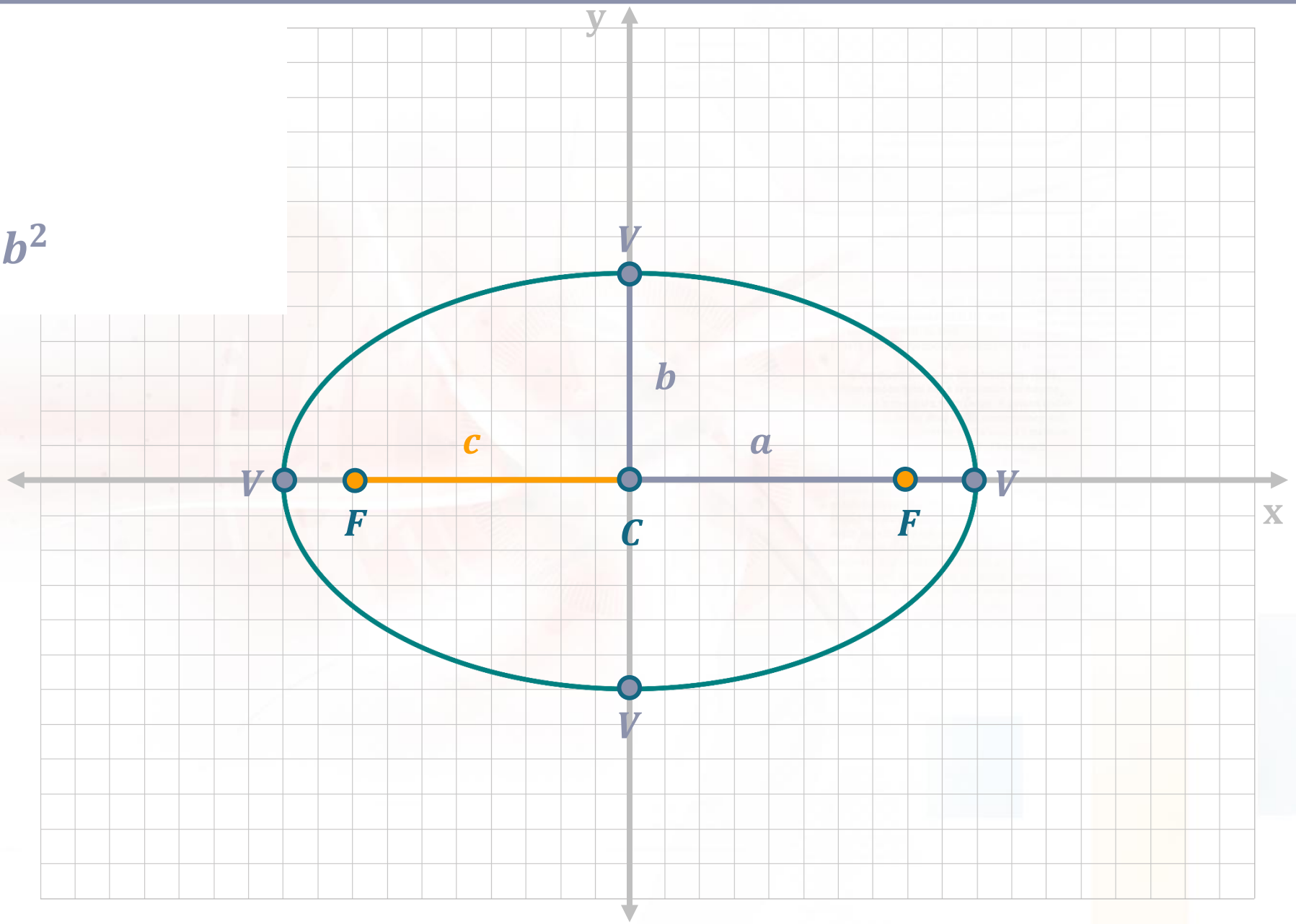
$2b$



foci

$F(\pm c, 0)$

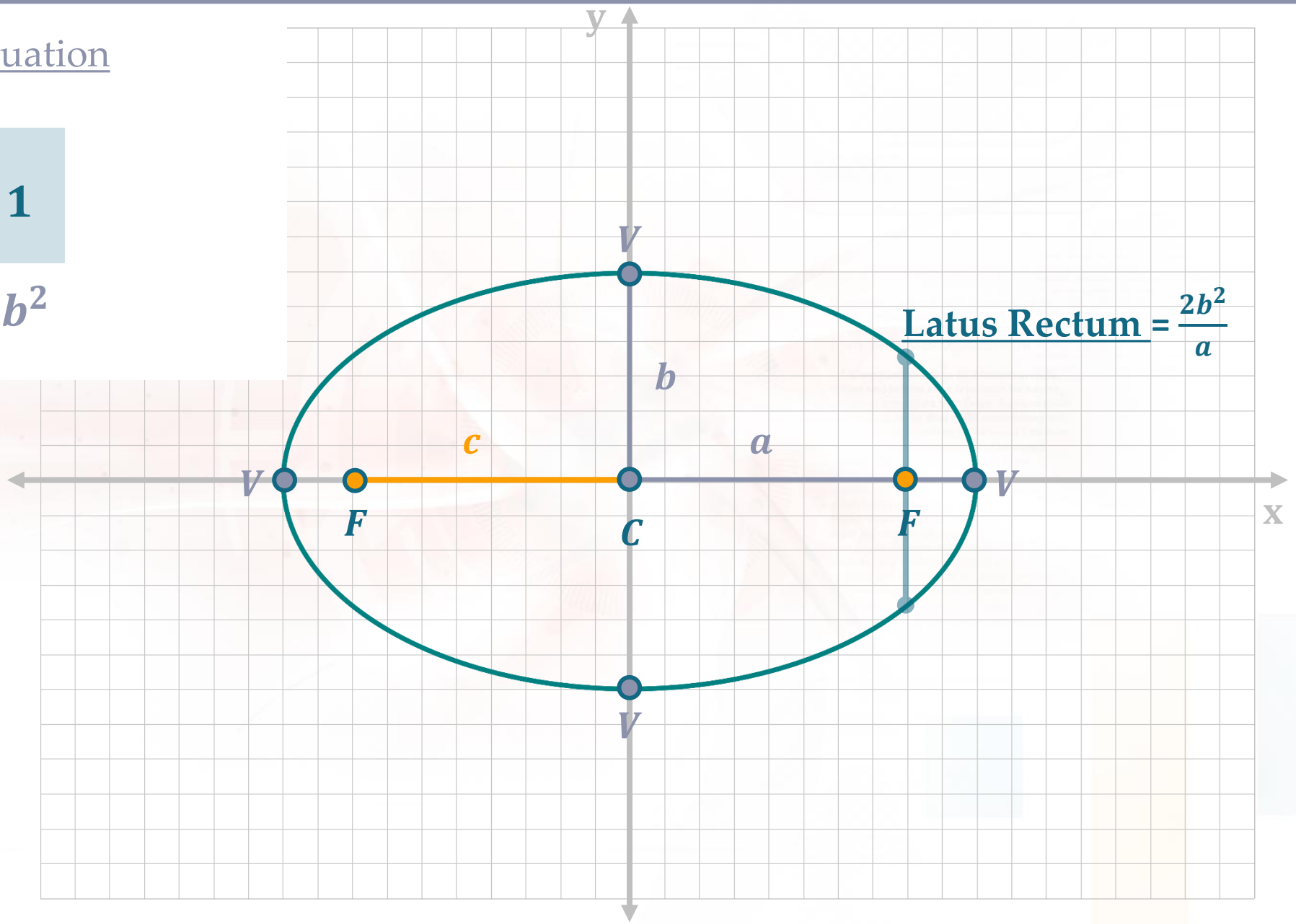
$c^2 = a^2 - b^2$



Standard Equation

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$$

$$c^2 = a^2 - b^2$$



MAJOR AXIS is on the $x - axis$

Standard Equation

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$$

$$c^2 = a^2 - b^2$$

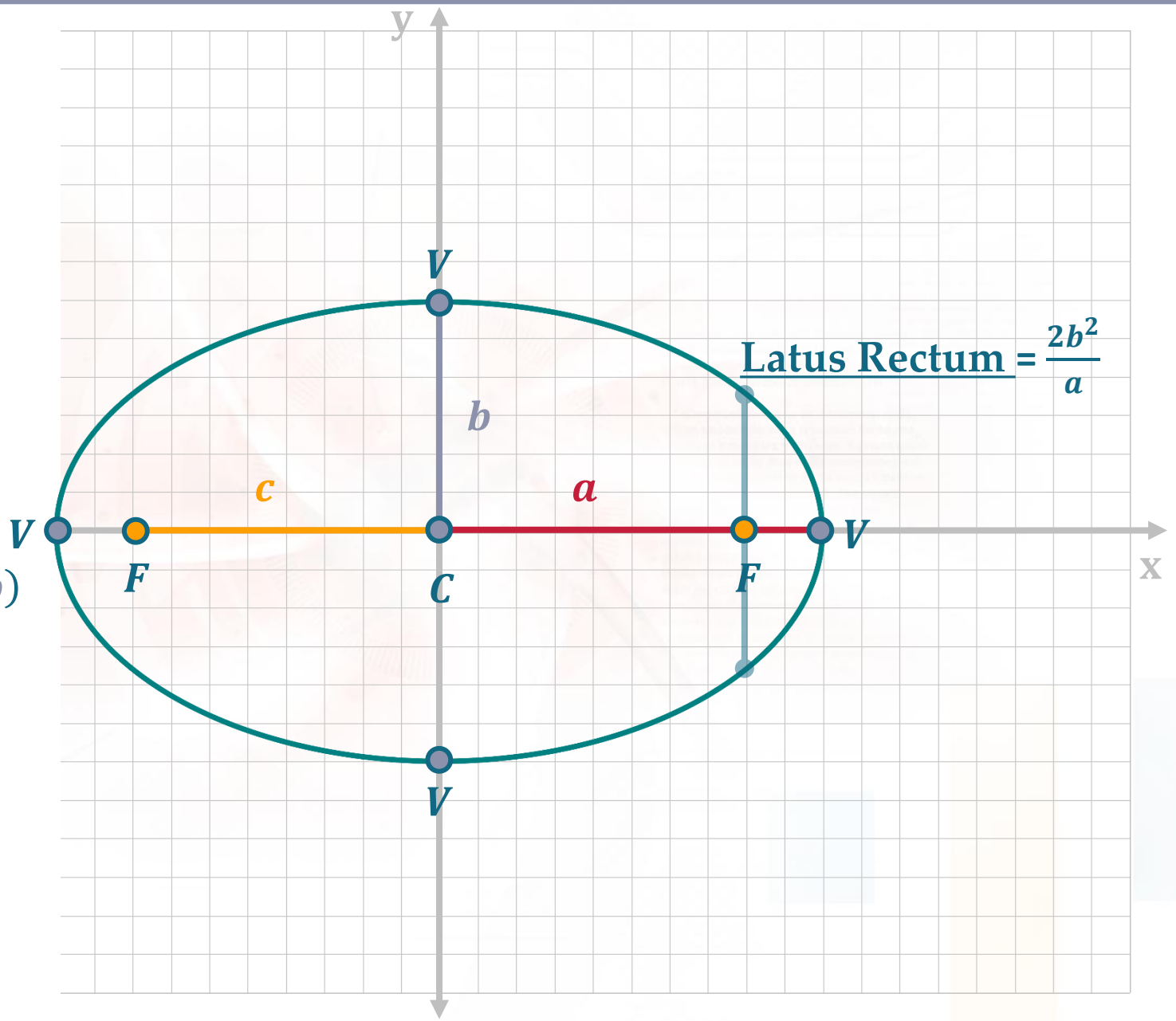
center, $C(0, 0)$

vertices, $V(\pm a, 0)$, $V(0, \pm b)$

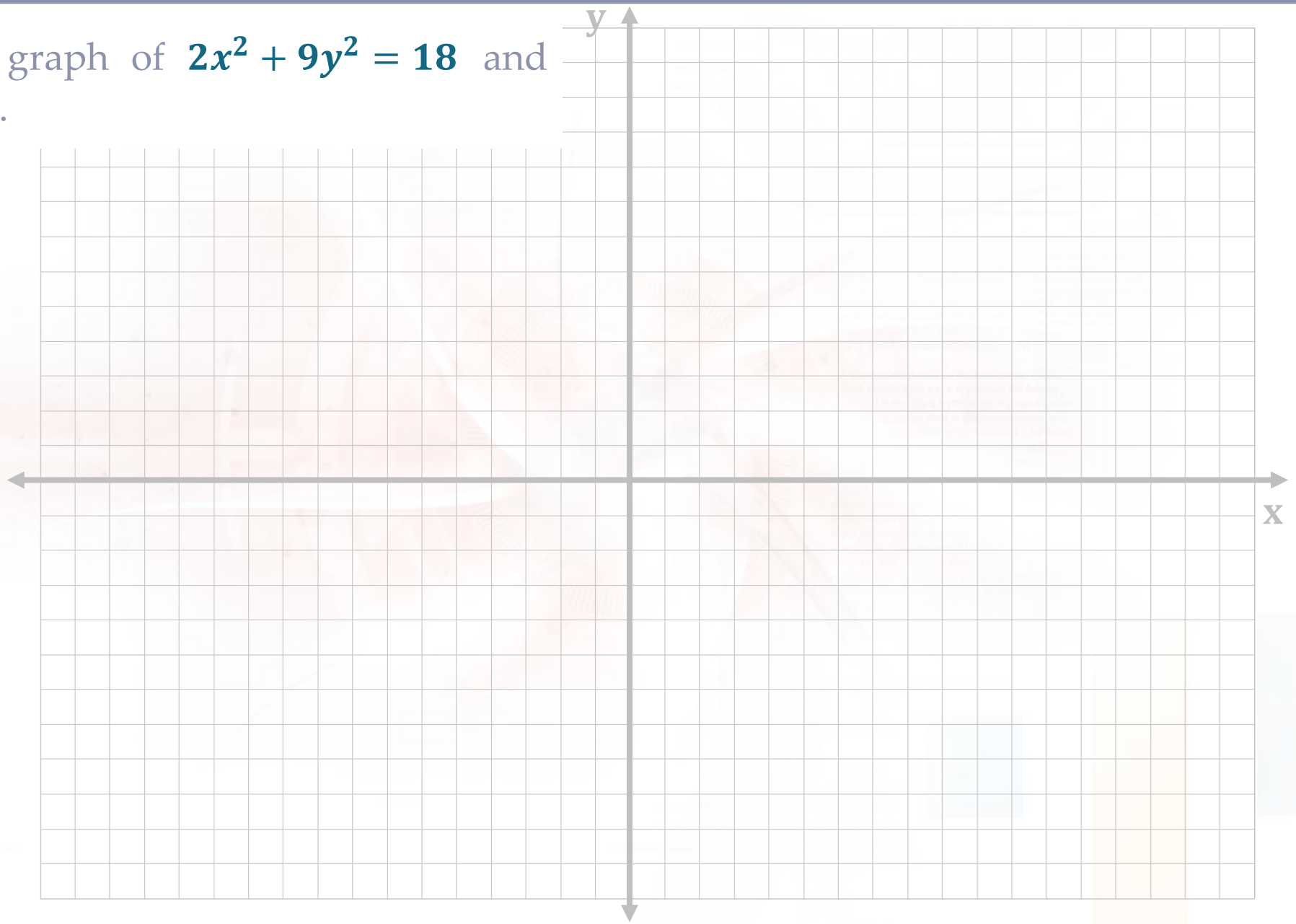
foci, $F(\pm c, 0)$

major axis, $2a$

minor axis, $2b$



Sketch the graph of $\mathbf{2x^2 + 9y^2 = 18}$ and find the **foci**.



MAJOR AXIS is on the $y - axis$

Standard Equation

$$\frac{x^2}{b^2} + \frac{y^2}{a^2} = 1$$

$$c^2 = a^2 - b^2$$

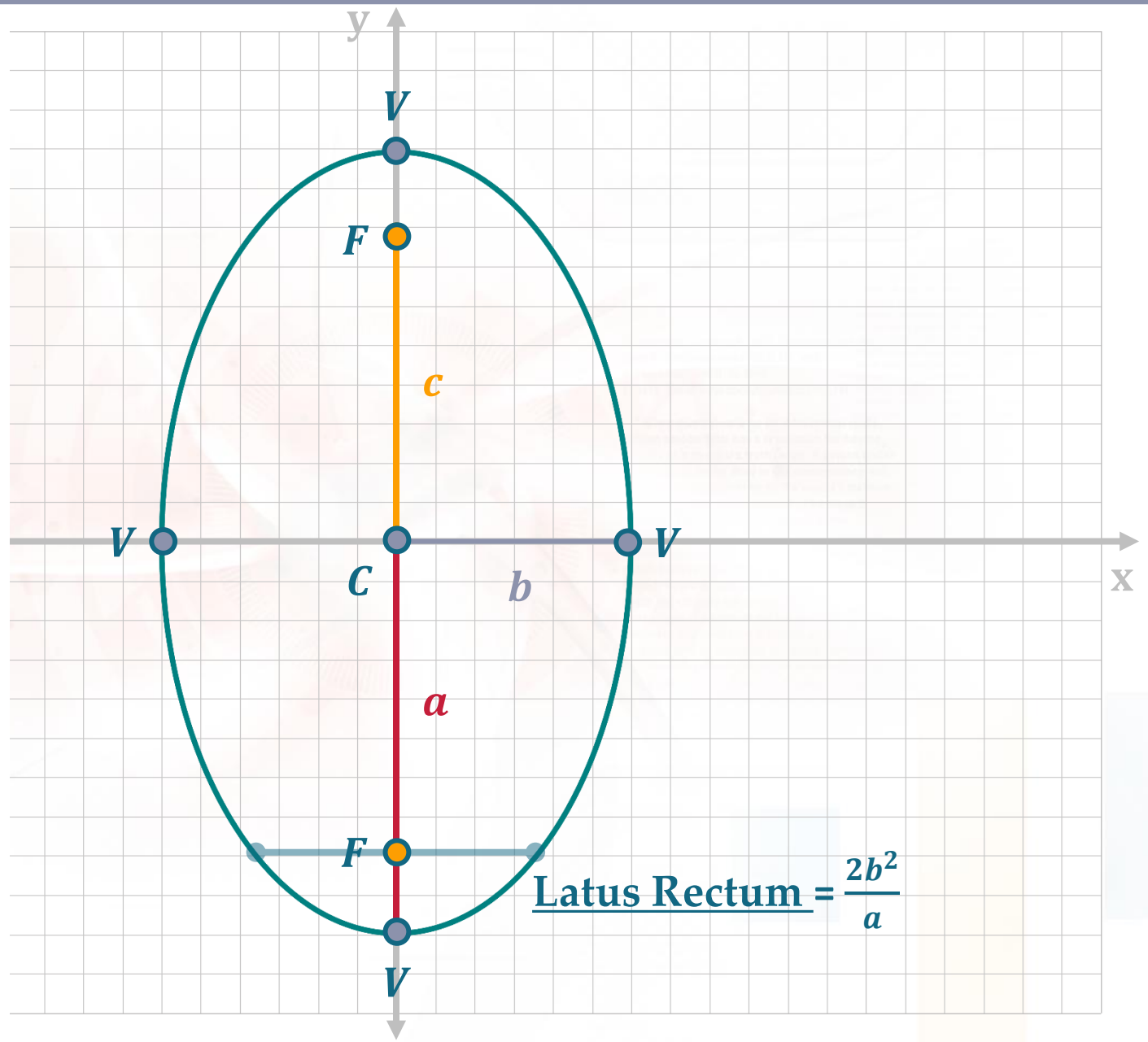
center, $C(0, 0)$

vertices, $V(\pm b, 0)$, $V(0, \pm a)$

foci, $F(0, \pm c)$

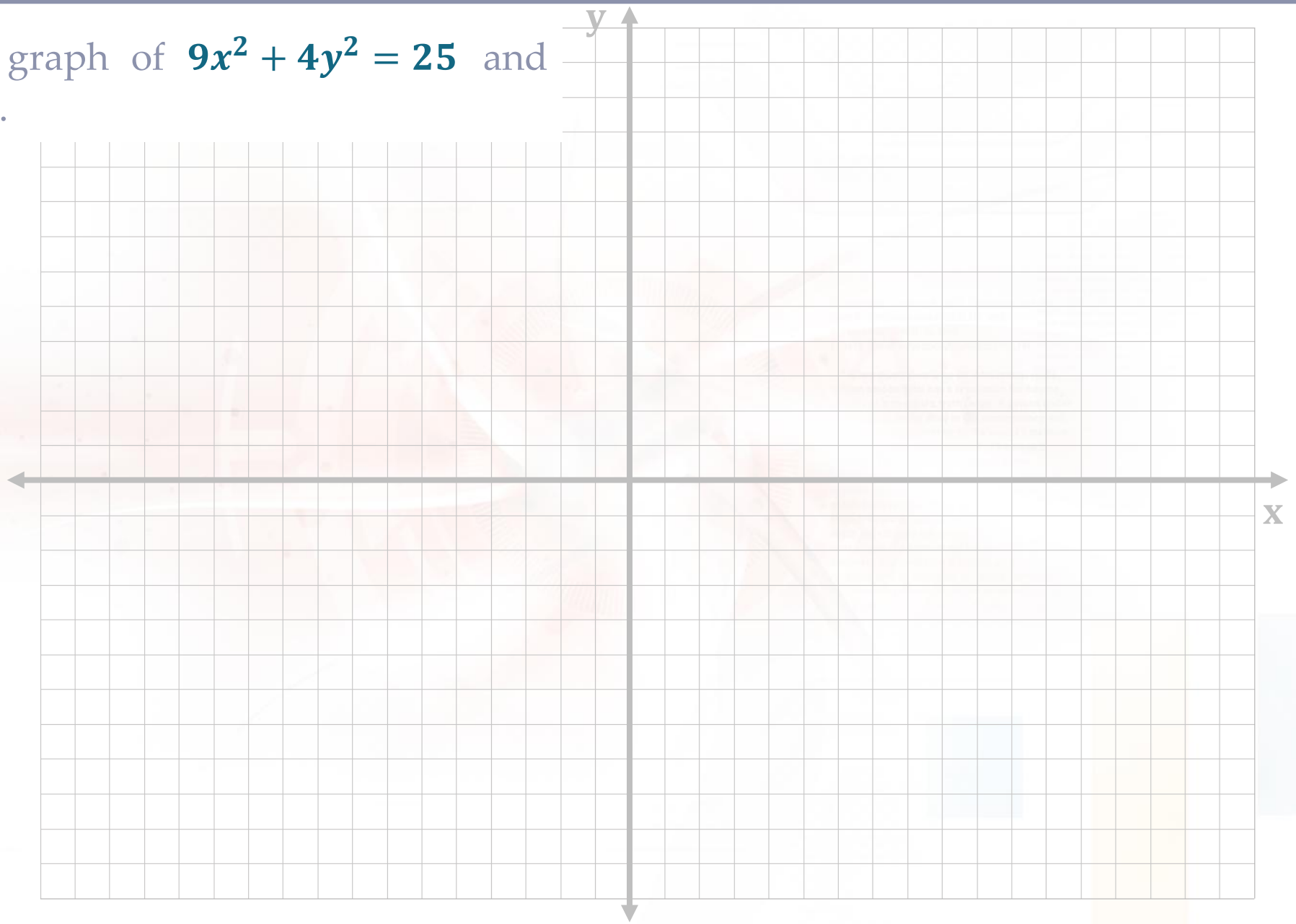
major axis, $2a$

minor axis, $2b$

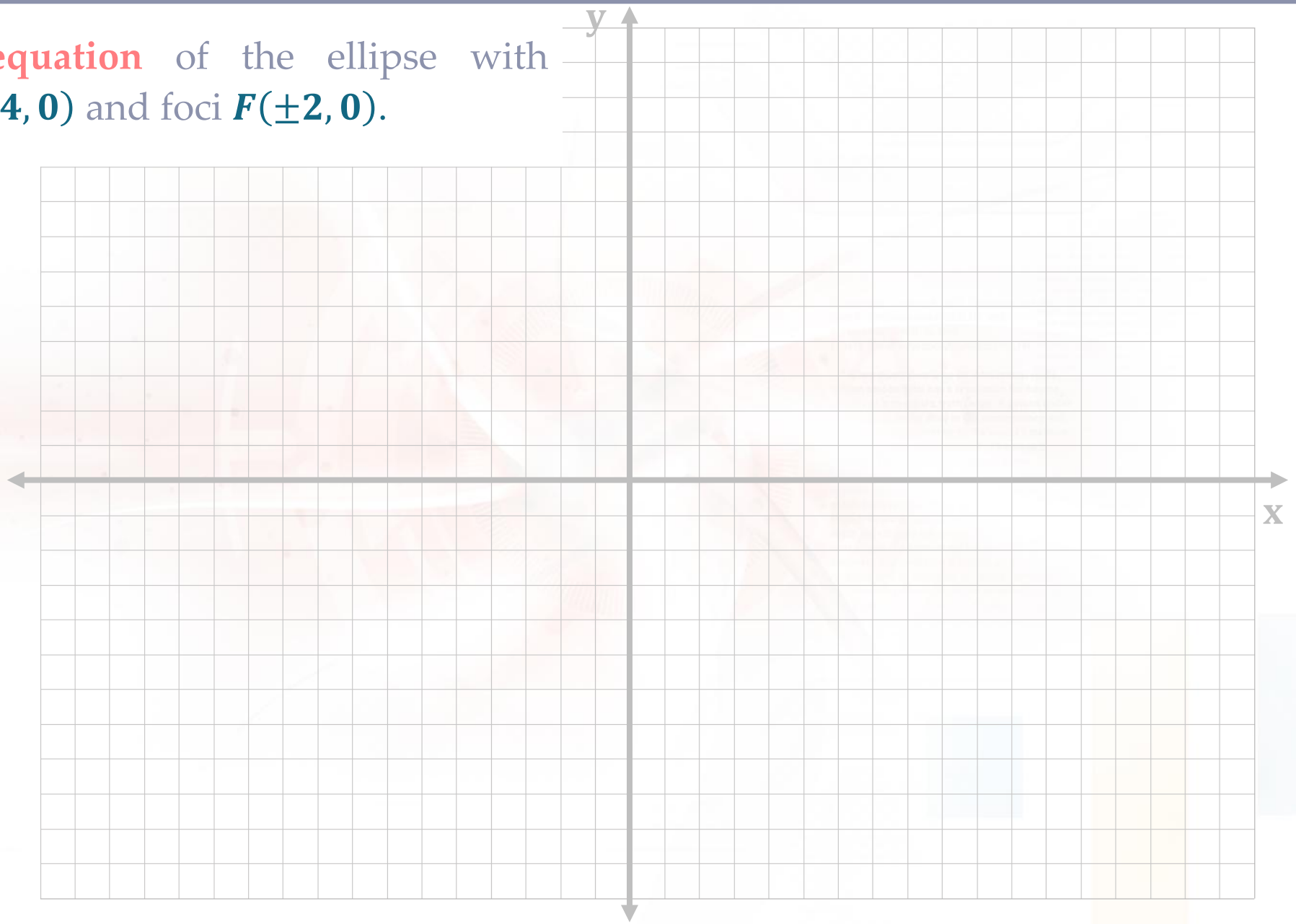


Latus Rectum = $\frac{2b^2}{a}$

Sketch the graph of $9x^2 + 4y^2 = 25$ and find the **foci**.



Find the **equation** of the ellipse with vertices $\boldsymbol{V(\pm 4, 0)}$ and foci $\boldsymbol{F(\pm 2, 0)}$.

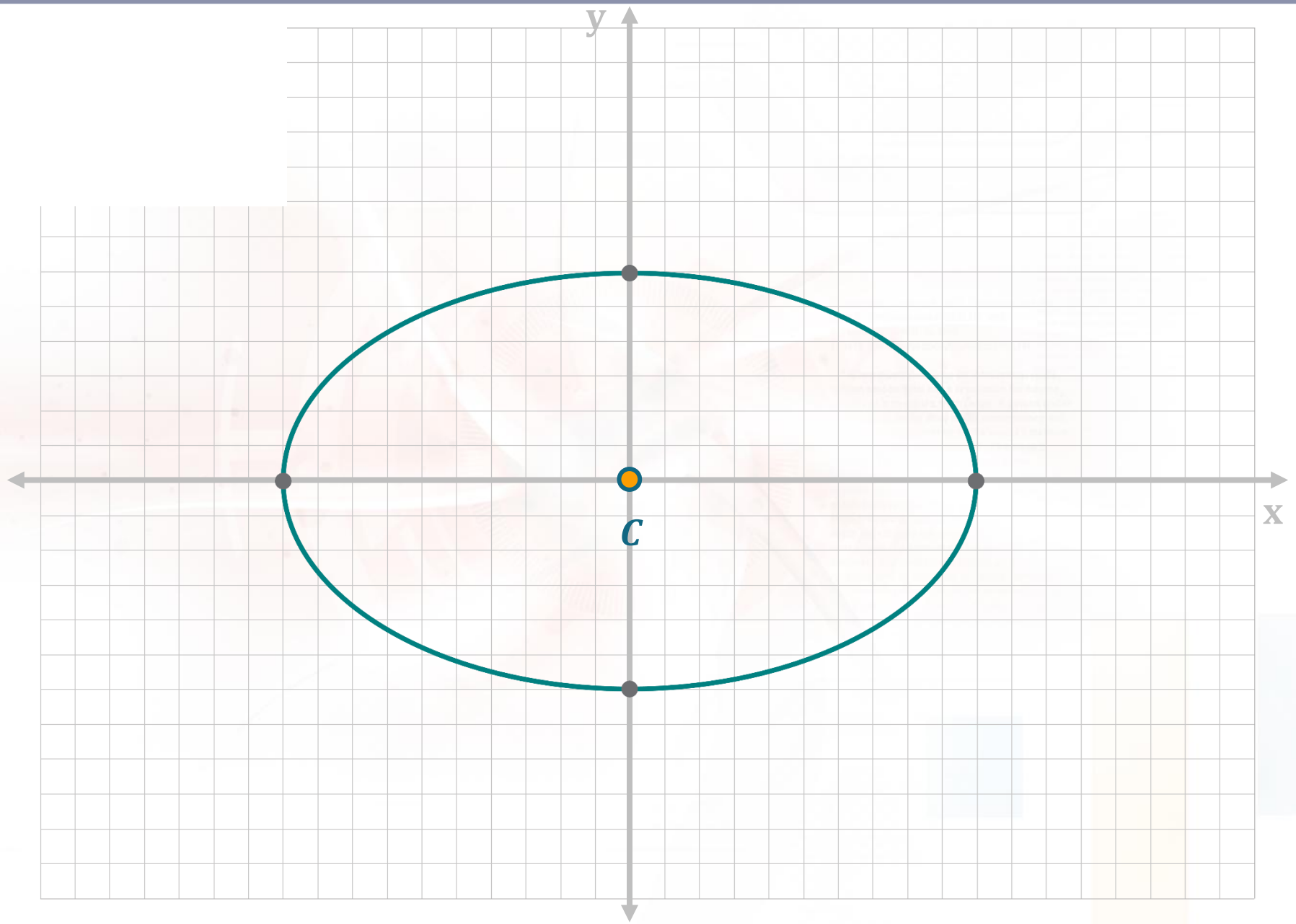


ELLIPSE WITH CENTER AT $C(h, k)$

CENTER AT THE ORIGIN

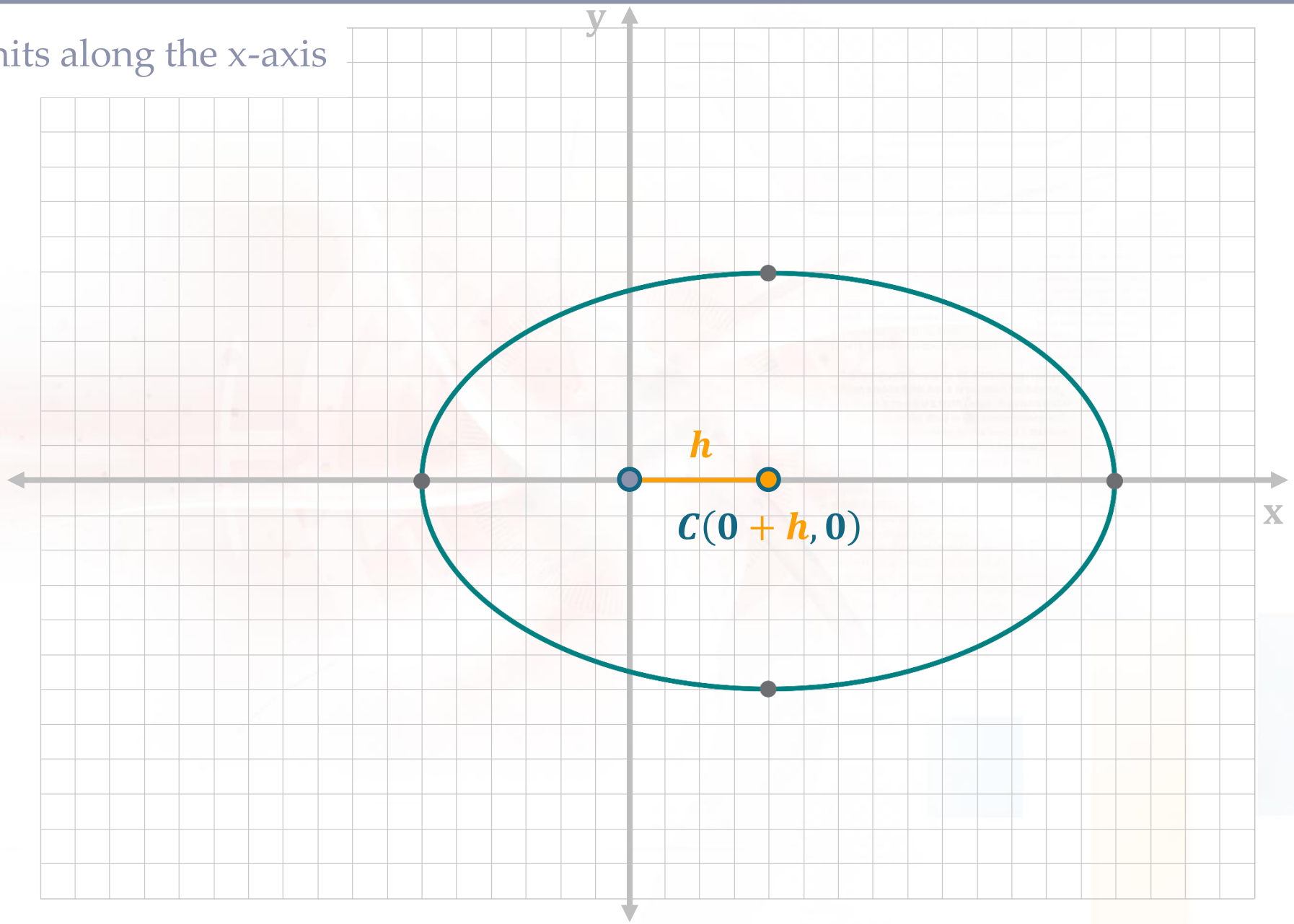
center

$C(0, 0)$



CENTER AT $C(h, 0)$

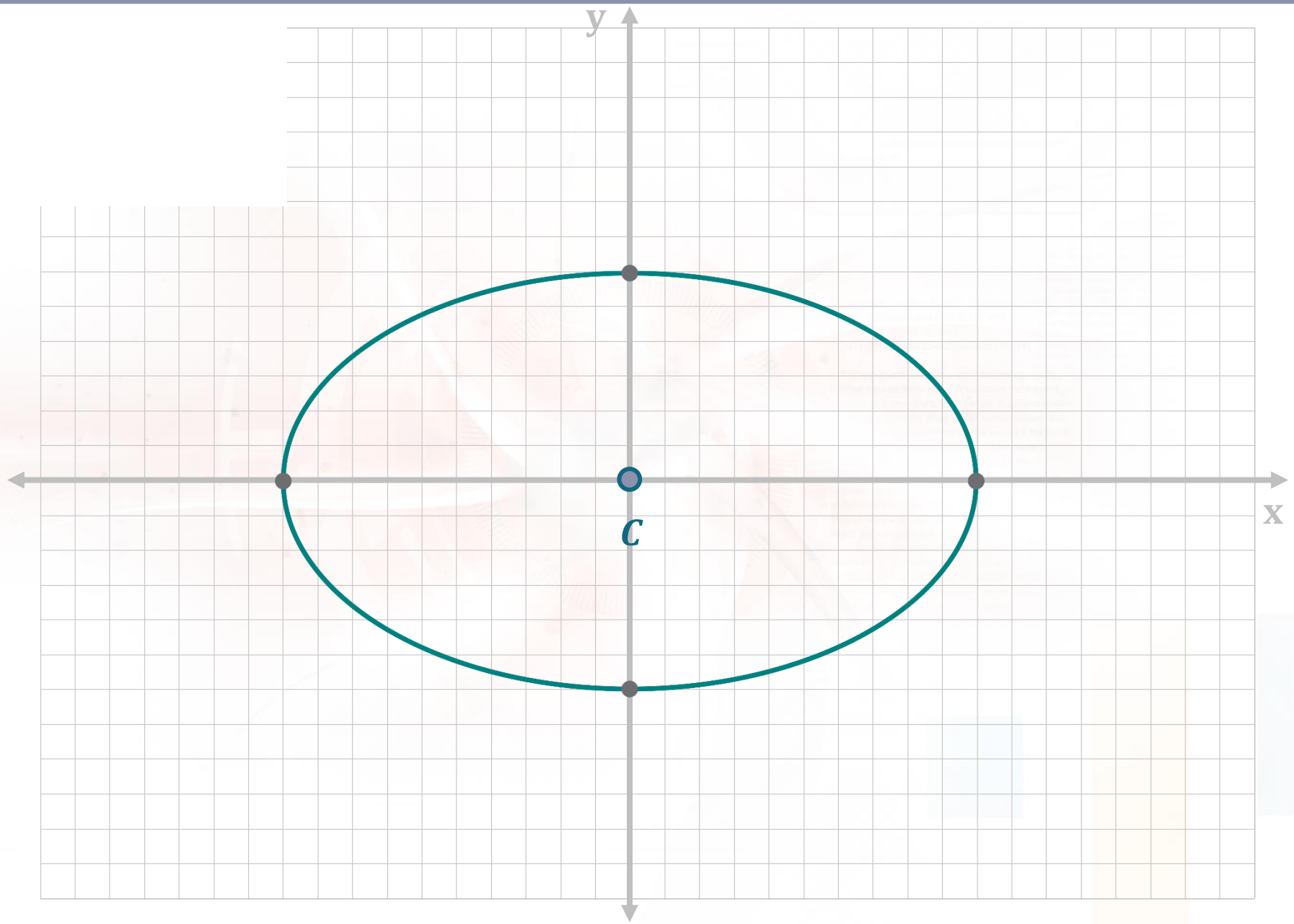
moving h units along the x-axis



CENTER AT THE ORIGIN

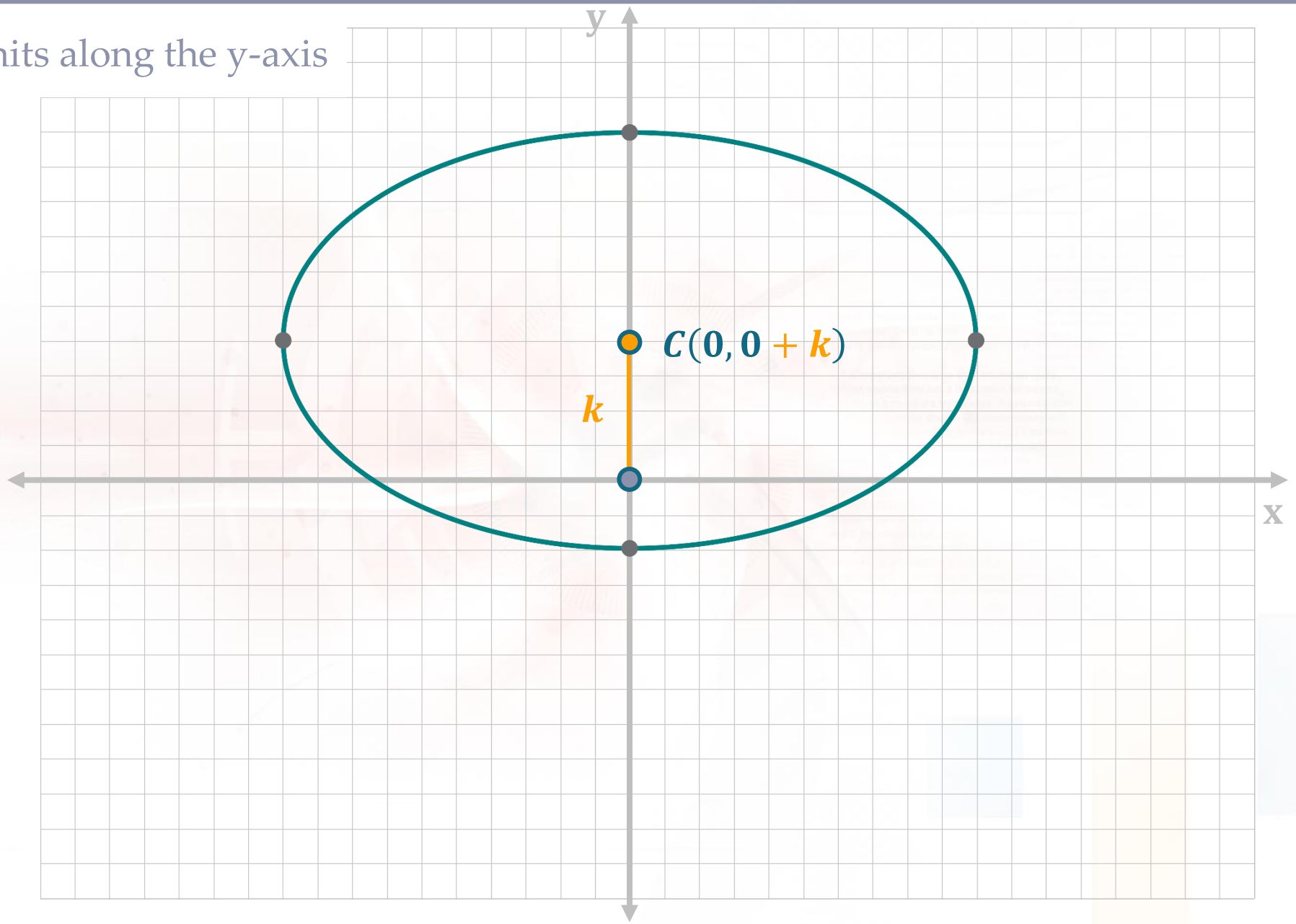
center

$C(0, 0)$



CENTER AT $C(0, k)$

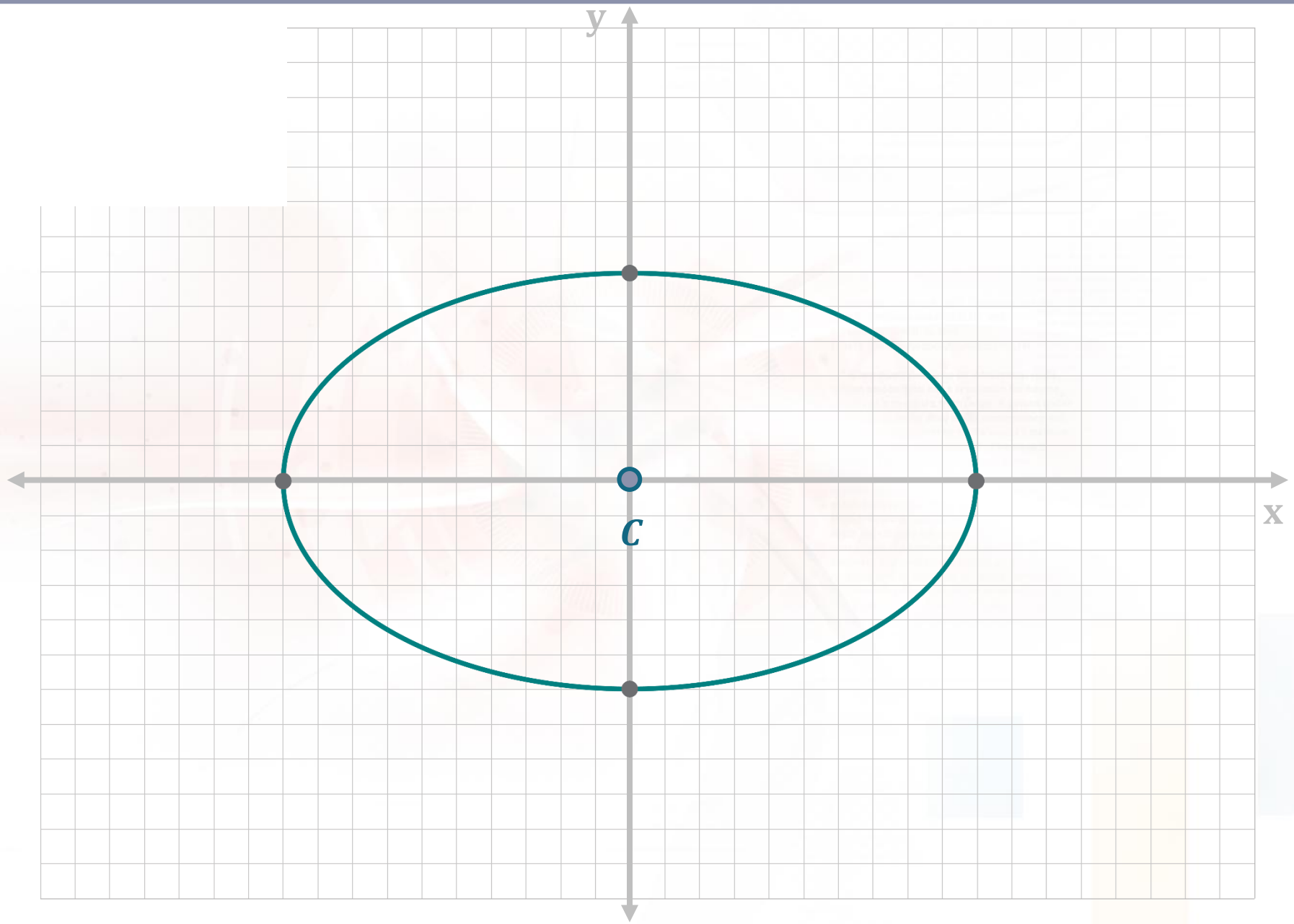
moving k units along the y-axis



CENTER AT THE ORIGIN

center

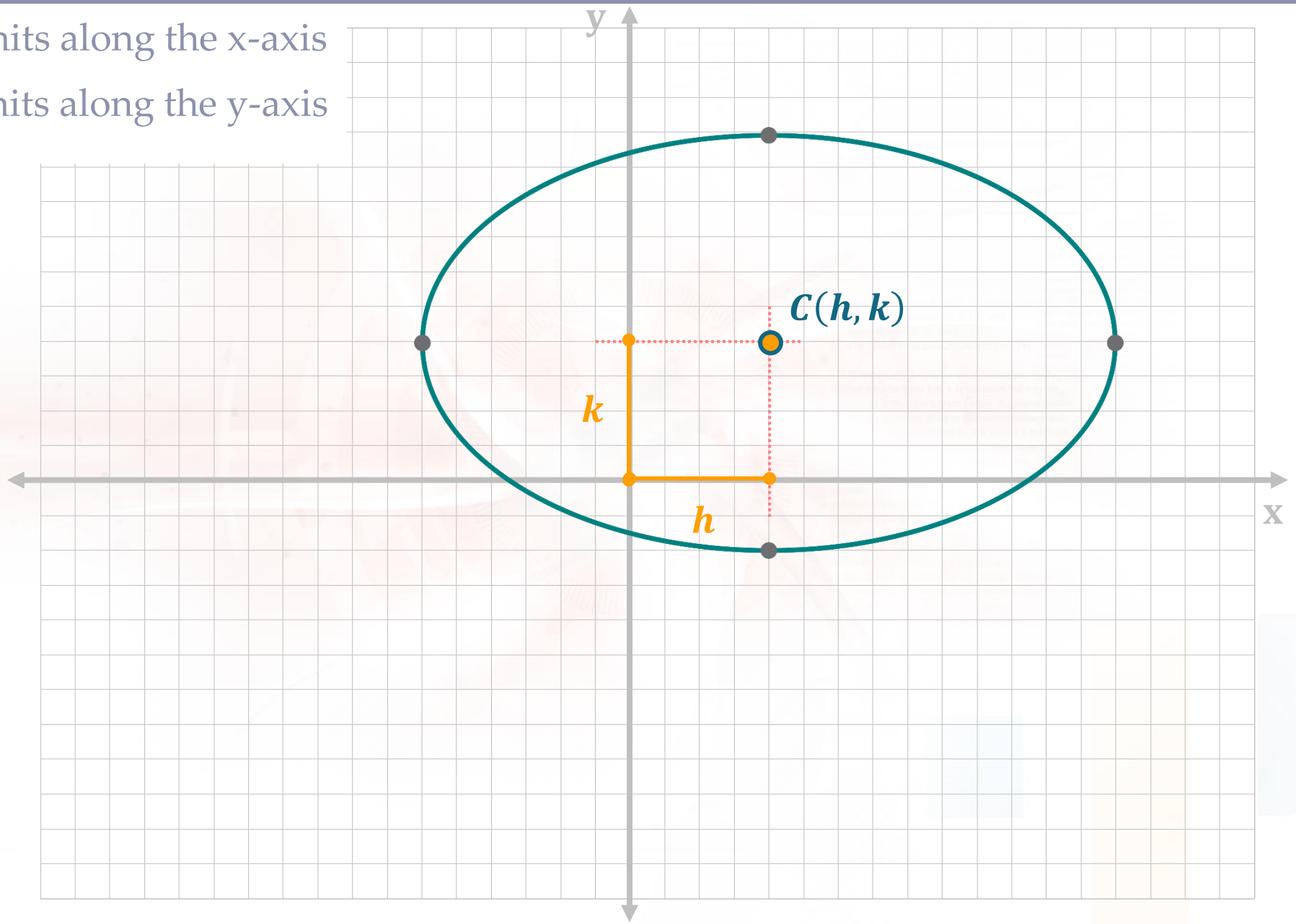
$C(0, 0)$



CENTER AT $C(h, k)$

moving h units along the x-axis

moving k units along the y-axis



MAJOR AXIS is along the $x - axis$

Standard Equation

$$\frac{(x-h)^2}{a^2} + \frac{(y-k)^2}{b^2} = 1$$

$$c^2 = a^2 - b^2$$

center, $C(h, k)$

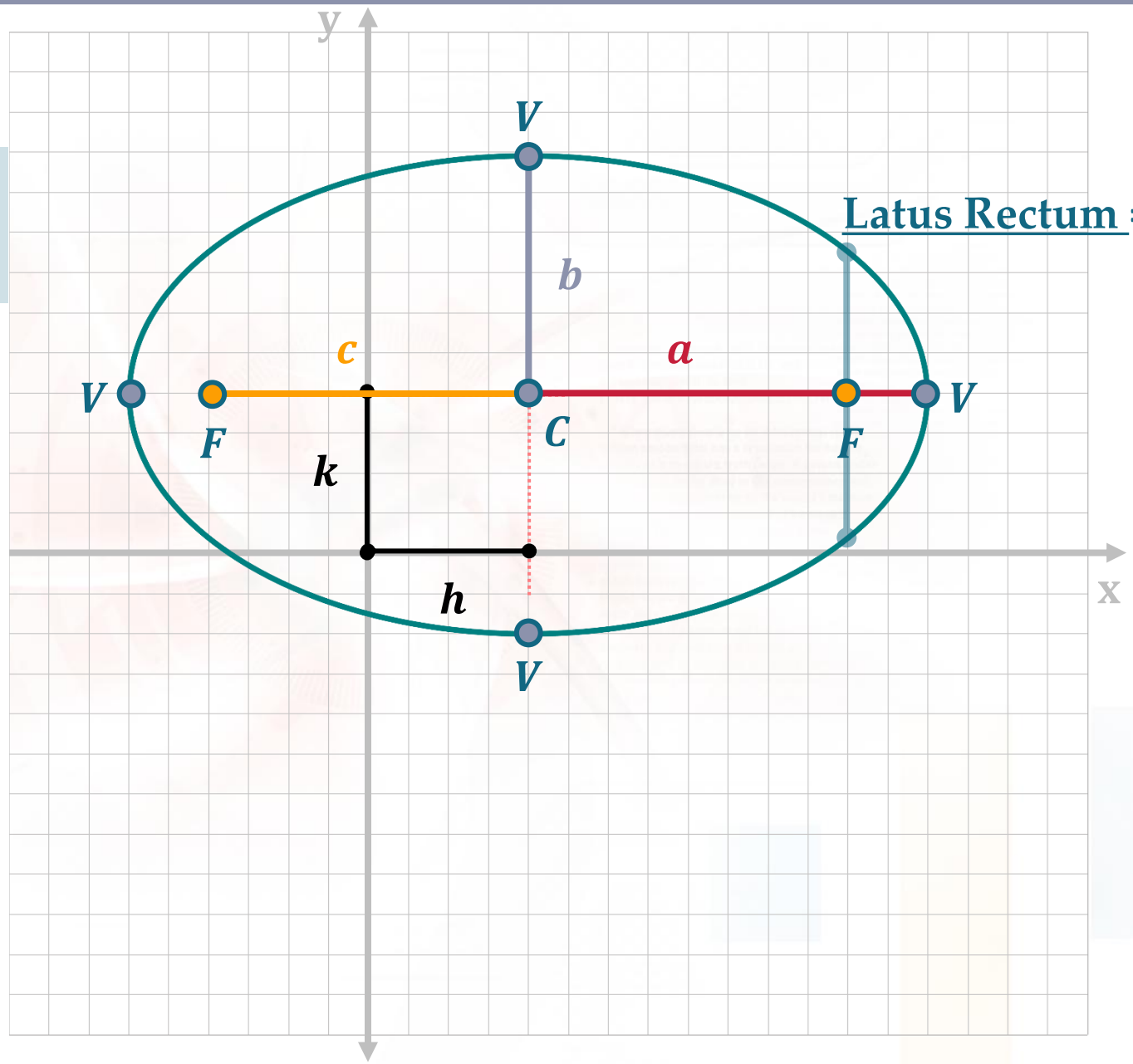
vertices, $V(h \pm a, k)$,

$V(h, k \pm b)$

foci, $F(h \pm c, k)$

major axis, $2a$

minor axis, $2b$



Latus Rectum = $\frac{2b^2}{a}$

MAJOR AXIS is along the $y - axis$

Standard Equation

$$\frac{(x-h)^2}{b^2} + \frac{(y-k)^2}{a^2} = 1$$

$$c^2 = a^2 - b^2$$

center, $C(h, k)$

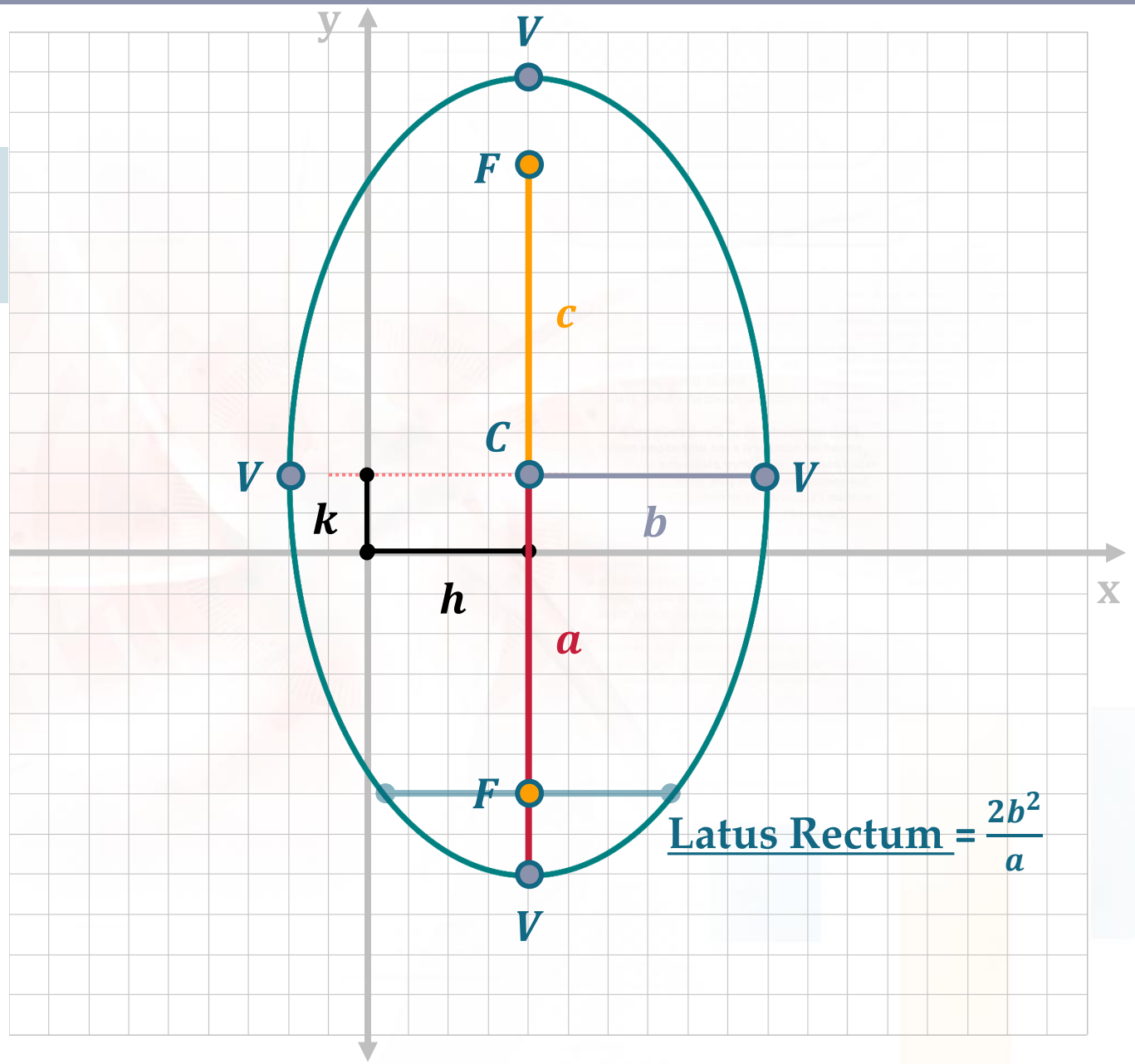
vertices, $V(h, k \pm a)$,

$V(h \pm b, k)$

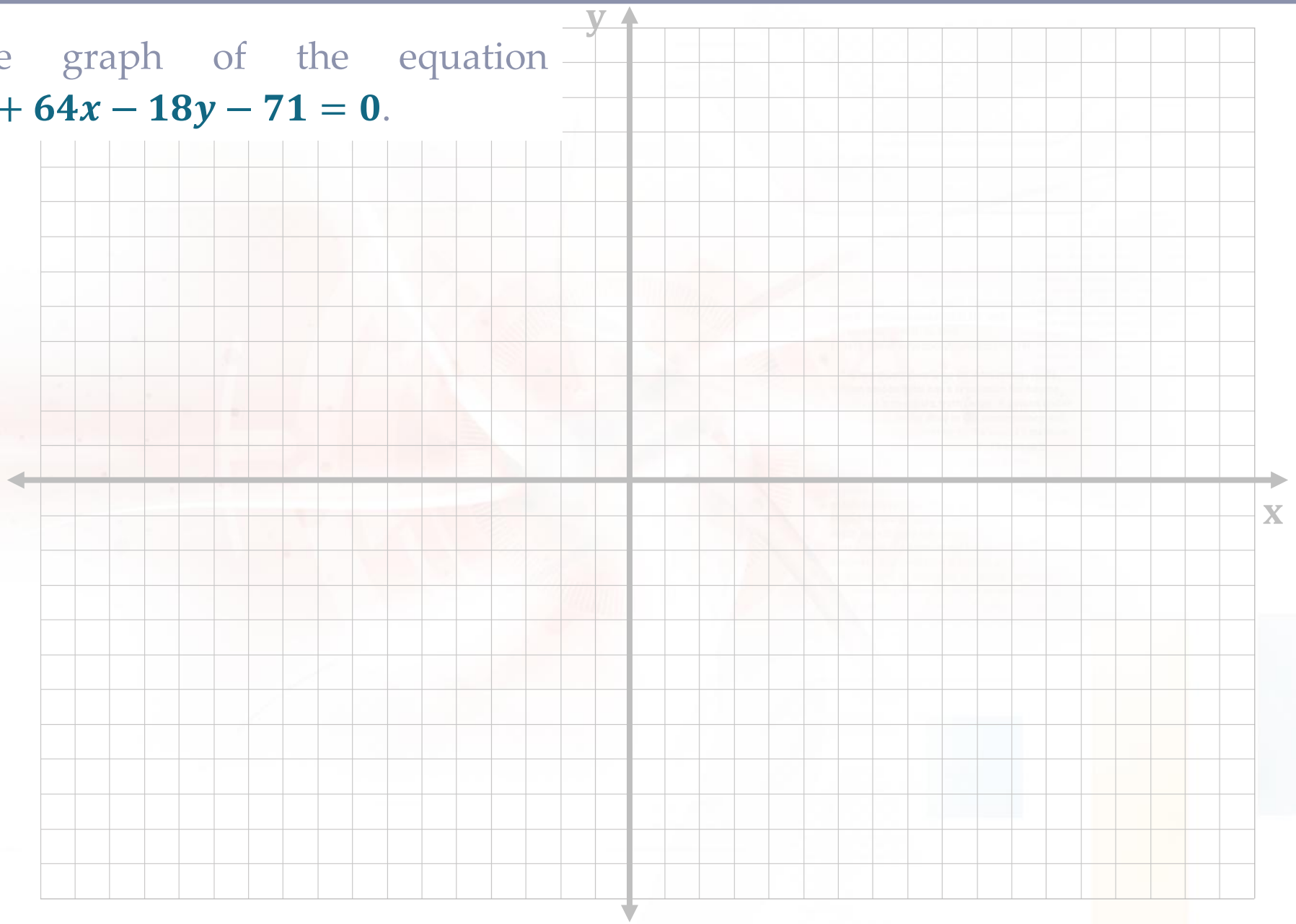
foci, $F(h, k \pm c)$

major axis, $2a$

minor axis, $2b$



Sketch the graph of the equation $\mathbf{16x^2 + 9y^2 + 64x - 18y - 71 = 0}$.



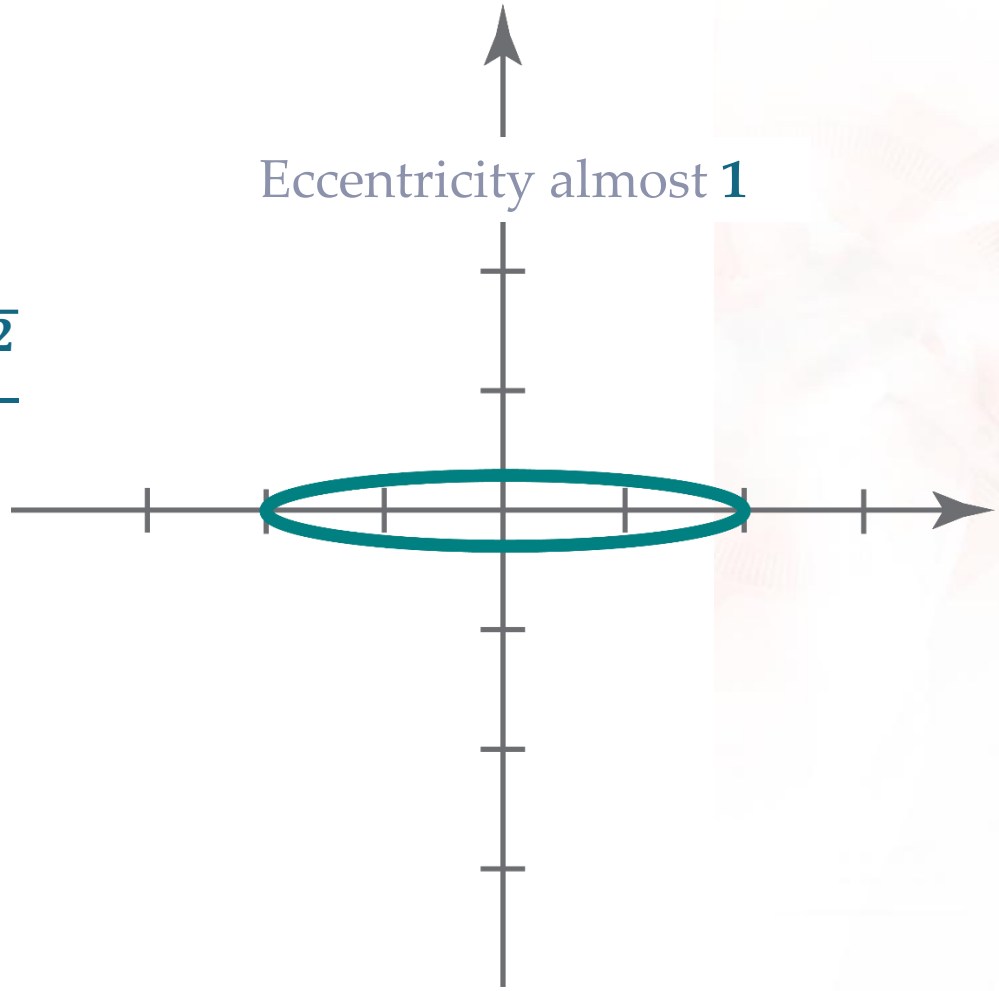
THE ECCENTRICITY OF AN ELLIPSE

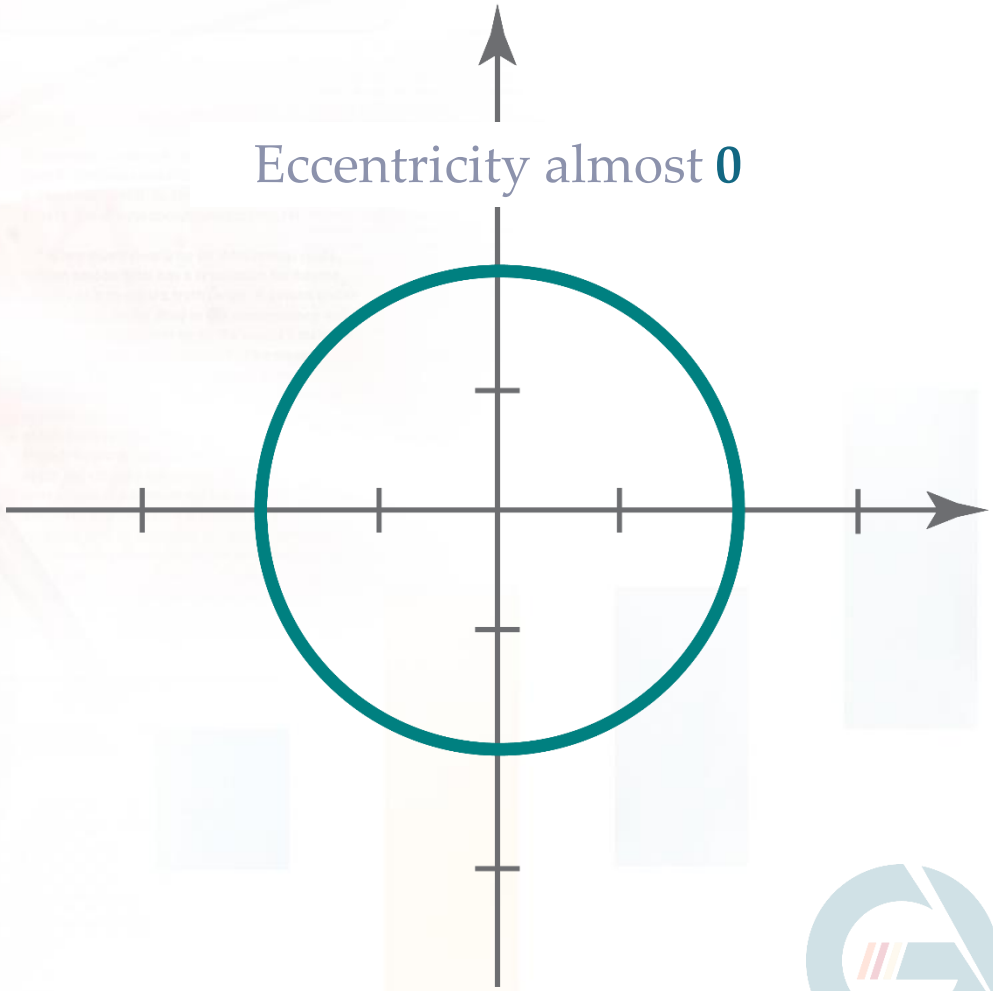
ECCENTRICITY, e

$$e = \frac{distance\ from\ the\ center\ to\ focus}{distance\ from\ the\ center\ to\ vertex}$$

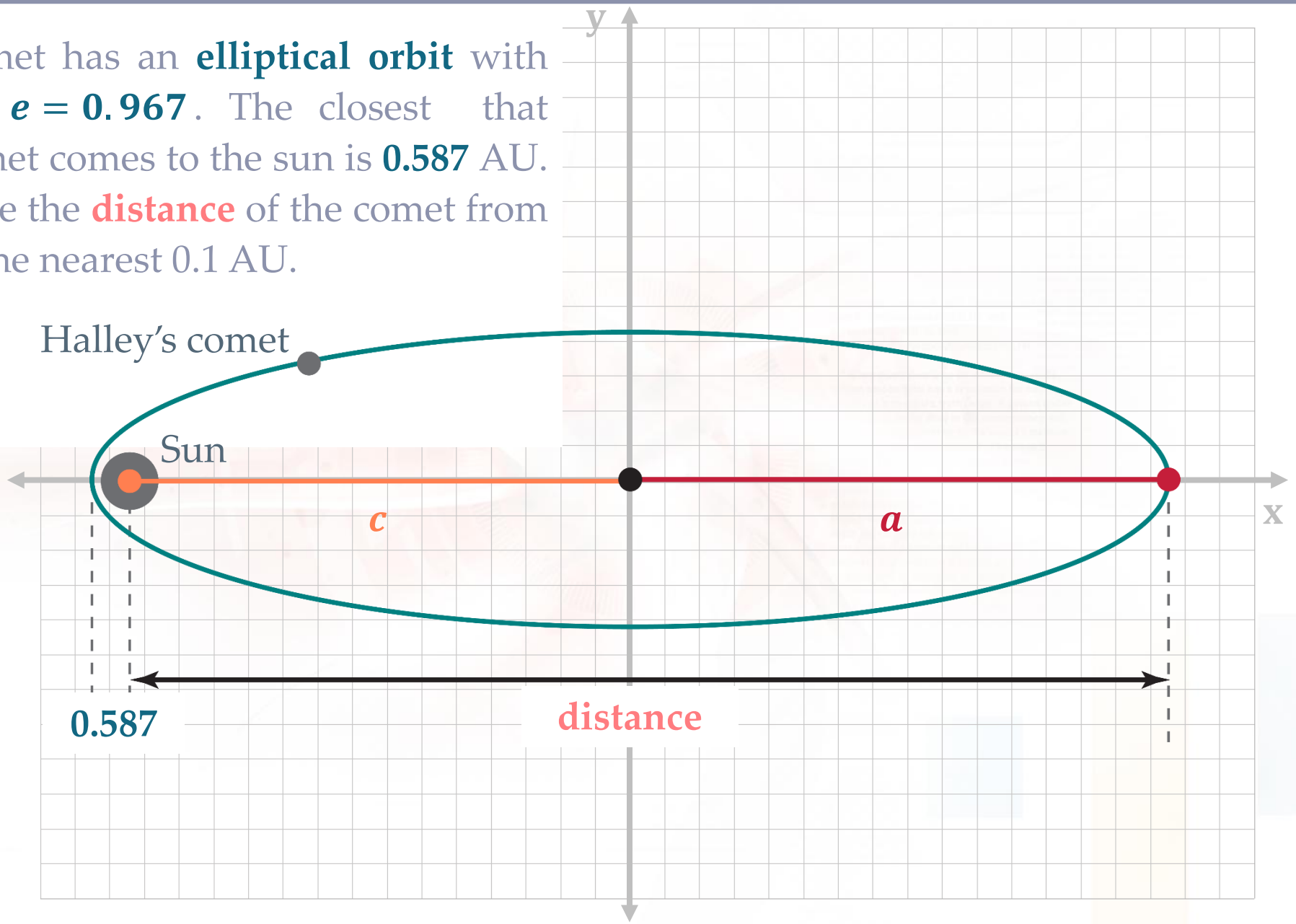
$$\boldsymbol{e = \frac{c}{a}}$$

$$\boldsymbol{e = \frac{\sqrt{a^2 - b^2}}{a}}$$





Halley's comet has an **elliptical orbit** with eccentricity $e = 0.967$. The closest that Halley's comet comes to the sun is **0.587** AU. Approximate the **distance** of the comet from the sun, to the nearest 0.1 AU.



SEATWORK